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// lab11.cpp : This file contains the 'main' function. Program execution begins and ends there.
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//
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#include <Windows.h>
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#include <iostream>
```

```
int main() {
    SYSTEM_INFO sysInfo;
    GetSystemInfo(&sysInfo);
    std::cout << "System page size: " << sysInfo.dwPageSize << " bytes" << std::endl;

    const SIZE_T memsize = 1 << 20;
    LPVOID memory = VirtualAlloc(NULL, memsize, MEM_COMMIT | MEM_RESERVE, PAGE_READWRITE);

    MEMORY_BASIC_INFORMATION memInfo;
    VirtualQuery(memory, &memInfo, sizeof(memInfo));
    printf("After Allocation: ");

    switch (memInfo.State) {
    case MEM_COMMIT:
        std::cout << "Committed" << std::endl;
        break;
    case MEM_RESERVE:
        std::cout << "Reserved" << std::endl;
        break;
    case MEM_FREE:
        std::cout << "Free" << std::endl;
        break;
    default:
        std::cout << "Unknown" << std::endl;
    }

    VirtualFree(memory, 0, MEM_RELEASE);
    VirtualQuery(memory, &memInfo, sizeof(memInfo));
    printf("After Deallocation: ");

    switch (memInfo.State) {
    case MEM_COMMIT:
        std::cout << "Committed" << std::endl;
        break;
    case MEM_RESERVE:
        std::cout << "Reserved" << std::endl;
        break;
    case MEM_FREE:
        std::cout << "Free" << std::endl;
        break;
    default:
        std::cout << "Unknown" << std::endl;
    }

    std::cin.get(); // Keeps console open
    return 0;
}
```

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// Run program: Ctrl + F5 or Debug > Start Without Debugging menu
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// Debug program: F5 or Debug > Start Debugging menu
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// Tips for Getting Started:
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// 1. Use the Solution Explorer window to add/manage files
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// 2. Use the Team Explorer window to connect to source control
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// 3. Use the Output window to see build output and other messages
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// 4. Use the Error List window to view errors
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// 5. Go to Project > Add New Item to create new code files, or Project > Add Existing Item to add existing code files to the project
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// 6. In the future, to open this project again, go to File > Open > Project and select the .sln file
```